

Last login: Tue Jun 2 19:53:50 on ttys003
adityadave@Adityas-MacBook-Air-3 ~ % psql -d postgres
psql (18.4)
Type "help" for help.

```
postgres=# \c tpch
You are now connected to database "tpch" as user "adityadave".
tpch=# SET ROLE alice;
WARNING: ===== RAW PARSE TREE =====
DETAIL:      {RAWTMT
:stmt
{VARIABLESETTMT
:kind 0
:name role
:args (
{A_CONST
:val "alice"
:location 9
}
)
:jumble_args false
:is_local false
:location 9
}
:stmt_location 0
:stmt_len 14
}
```

WARNING: ===== ANALYZED QUERY =====

```
DETAIL:      {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VARIABLESETTMT
:kind 0
:name role
:args (
{A_CONST
:val "alice"
:location 9
}
)
:jumble_args false
:is_local false
:location 9
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 14
}
```

```
SET
tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT
```

```
o_orderpriority,
COUNT(*) AS order_count
FROM orders
WHERE o_orderdate >= DATE '1994-01-01'
AND o_orderdate < DATE '1994-01-01' + INTERVAL '3' MONTH
AND EXISTS (
SELECT 1
FROM lineitem
WHERE l_orderkey = o_orderkey
AND l_commitdate < l_receiptdate
)
```

```
GROUP BY o_orderpriority
ORDER BY o_orderpriority;
WARNING: ===== RAW PARSE TREE =====
DETAIL:      {RAWTMT
```

```
{stmt
{EXPLAINSTMT
:query
{SELECTTMT
:distinctClause <>
:intoClause <>
:targetList (
{REXTARGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields ("o_orderpriority")
:location 47
}
:location 47
}
{REXTARGET
:name order_count
:indirection <>
:val
{FUNCCALL
:funcname ("count")
:args <>
:agg_order <>
:agg_filter <>
:over <>
:agg_within_group false
:agg_star true
:agg_distinct false
:func_variadic false
:funcformat 0
:location 68
}
:location 68
}
}
:fromClause (
{RANGEVAR
:catalogname <>
```

```

:schemaname <>
:relname orders
:inh true
:relpersistance p
:alias <>
:location 97
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name (">=")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 110
}
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 130
}
}
:typeName
{TYPENAME
:names ("date")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 125
}
:location -1
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 122
}
{A_EXPR
:name ("<")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 149
}
}
:rexpr
{A_EXPR
:name ("*")
:lexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 168
}
}
:typeName
{TYPENAME
:names ("date")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 163
}
:location -1
}
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\3"
:location 192
}
}
:typeName
{TYPENAME
:names ("pg_catalog" "interval")
:typeOid 0
:setof false
:pct_type false
:typmods (
{A_CONST
:val 2
:location 196
}
)
:typemod -1
:arrayBounds <>
:location 183
}
:location -1
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 181
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 161
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTARTGET
:name <>
:indirection <>
:val
{A_CONST
:val 1
:location 232
}
}
:location 232
}
}
}
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname lineitem
:inh true
:relpersistance p
:alias <>
:location 247
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("l_orderkey")
:location 270
}
}
)
}

```

```

    }
    :rexpr
    {COLUMNREF
     :fields ("o_orderkey")
     :location 283
    }
    :rexpr_list_start 0
    :rexpr_list_end 0
    :location 281
  }
  {A_EXPR
  :name ("<")
  :lexpr
  {COLUMNREF
   :fields ("l_commitdate")
   :location 308
  }
  :rexpr
  {COLUMNREF
   :fields ("l_receiptdate")
   :location 323
  }
  :rexpr_list_start 0
  :rexpr_list_end 0
  :location 321
  }
  )
  :location 304
  }
  :groupClause <>
  :groupDistinct false
  :havingClause <>
  :windowClause <>
  :valueLists <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :lockingClause <>
  :withClause <>
  :op 0
  :all false
  :larg <>
  :rarg <>
  }
  :location 208
  }
  )
  :location 145
  }
  :groupClause (
  {COLUMNREF
   :fields ("o_orderpriority")
   :location 350
  }
  )
  :groupDistinct false
  :havingClause <>
  :windowClause <>
  :valueLists <>
  :sortClause (
  {SORTBY
   :node
   {COLUMNREF
    :fields ("o_orderpriority")
    :location 375
   }
   :sortby_dir 0
   :sortby_nulls 0
   :useOp <>
   :location -1
  }
  )
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :lockingClause <>
  :withClause <>
  :op 0
  :all false
  :larg <>
  :rarg <>
  }
  :options (
  {DEFELEM
   :defnamespace <>
   :defname analyze
   :arg <>
   :defaction 0
   :location 9
  }
  {DEFELEM
   :defnamespace <>
   :defname verbose
   :arg <>
   :defaction 0
   :location 18
  }
  {DEFELEM
   :defnamespace <>
   :defname buffers
   :arg <>
   :defaction 0
   :location 27
  }
  )
  )
  :stmt_location 0
  :stmt_len 390
  }

```

WARNING: ===== ANALYZED QUERY =====

```

DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{EXPLAINSTMT
:query
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalpri
ce" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority"
"o_comment")
}
}
:rtekind 0

```

```

:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:varitypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCI false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 110
}
){CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 130
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 122
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 149
}
){OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 168
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
){CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 192
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 0 ]
}
)
:location 181
}
)
:location 161

```

```

}
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_
linenumber" "l_quantity" "l_extendedprice" "l_disco
unt" "l_tax" "l_returnflag" "l_linestatus" "l_shipd
ate" "l_commitdate" "l_receiptdate" "l_shipinstruct
" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 19 20)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 270
}
)
:location 281
}
)
{OEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 308
}
)
:location 321
}
)
:location 304
}

```

```

    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
      {TARGETENTRY
        :expr
        {CONST
          :consttype 23
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location 232
          :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
  }
  :location 208
}
}
:location 145
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 1
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16561
    :resorigcol 0
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes <>
      :aggdirectargs <>
      :args <>
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggssplit 0
      :aggno -1
      :aggtransno -1
      :location 68
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>

```

```

:stmt_location 0
:stmt_len 0
}
:options (
{DEFELEM
:dfnamespace <>
:dfname analyze
:arg <>
:dfaction 0
:location 9
}
{DEFELEM
:dfnamespace <>
:dfname verbose
:arg <>
:dfaction 0
:location 18
}
{DEFELEM
:dfnamespace <>
:dfname buffers
:arg <>
:dfaction 0
:location 27
}
)
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperinfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 390
}

```

WARNING: ===== REWRITE INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:varitypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
}

```

```

:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 110
}
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 130
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 122
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 149
}
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 168
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:conststypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 192
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 181
}
)
:location 161
}
{SUBLINK
:sublinkid 0
:restexpr <>
:opertname <>
:subselect
{QUERY
:commandtype 1
:querysource 0
:canSetTag true
:utilitystmt <>
:resultrelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtkind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
}

```



```

:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16566
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 19 20)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
  )
  :quals
  {BOOLEXP
  :boolop and
  :args (
    {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 270
      }
      {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 1
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 283
      }
    )
    :location 281
  }
  {OPEXP
  :opno 1095
  :opfuncid 1087
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 12
    :location 308
    }
    {VAR
    :varno 1
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 13
    :location 323
    }
  )
  :location 321
  }
  :location 304
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
  :expr
  {CONST
  :consttype 23
  :consttypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull false
  :location 232
  :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
  }
  :resno 1
  :resname ?column?
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 208
}

```

```

    )
    :location 145
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 1
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16561
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes <>
      :aggdirectargs <>
      :args <>
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggsplit 0
      :aggnosyn -1
      :aggtransno -1
      :location 68
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: Table OID=16561
WARNING: ===== RLS POLICY ADDED =====
DETAIL: (
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
    {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStat <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSublinks false
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false
      :hasRowSecurity false
      :hasGroupRTE false
      :isReturn false
      :ctelist <>
      :rtable (
        {RANGETBLENTY
          :alias
          {ALIAS
            :aliasname c
            :colnames <>
          }
          :eref
          {ALIAS
            :aliasname c
            :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone"
              "c_acctbal" "c_mktsegment" "c_comment")
          }
          :rtekind 0
          :relid 16546
          :inh true
          :relkind r
          :rellockmode 1
        }
      )
    }
  }
)

```

```

:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:varitypmmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:varitypmmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:varitypmmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:constitypmmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]
}
}
:location -1
}
}
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:exp
{CONST
:consttype 23
:constitypmmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>

```

```

:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)

```

WARNING: ===== REWRITE OUTPUT =====

DETAIL: (

{QUERY

:commandType 1

:querySource 0

:canSetTag true

:utilityStmt <>

:resultRelation 0

:hasAggs true

:hasWindowFuncs false

:hasTargetSRFs false

:hasSubLinks true

:hasDistinctOn false

:hasRecursive false

:hasModifyingCTE false

:hasForUpdate false

:hasRowSecurity true

:hasGroupRTE true

:isReturn false

:ctelist <>

:rtable (

{RANGETBENTRY

:alias <>

:eref

{ALIAS

:aliasname orders

:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")

}

:rtekind 0

:relid 16561

:inh true

:relkind r

:rellockmode 1

:perminfoindex 1

:tablesample <>

:lateral false

:inFromCl true

:securityQuals (

{SUBLINK

:sublinkType 0

:sublinkId 0

:testexpr <>

:operName <>

:subselect

{QUERY

:commandType 1

:querySource 0

:canSetTag true

:utilityStmt <>

:resultRelation 0

:hasAggs false

:hasWindowFuncs false

:hasTargetSRFs false

:hasSubLinks false

:hasDistinctOn false

:hasRecursive false

:hasModifyingCTE false

:hasForUpdate false

:hasRowSecurity false

:hasGroupRTE false

:isReturn false

:ctelist <>

:rtable (

{RANGETBENTRY

:alias

{ALIAS

:aliasname c

:colnames <>

}

:eref

{ALIAS

:aliasname c

:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")

}

:rtekind 0

:relid 16546

:inh true

:relkind r

:rellockmode 1

:perminfoindex 1

:tablesample <>

:lateral false

:inFromCl true

:securityQuals <>

}

)

:rteperminfos (

{RTEPERMISSIONINFO

:relid 16546

:inh true

:requiredPerms 2

:checkAsUser 0

:selectedCols (b 8 13)

:insertedCols (b)

:updatedCols (b)

}

)

:jointree

{FROMEXPR

:fromlist (

{RANGETBLREF

:rtindex 1

}

)

:quals

{BOOLEXP

:boolop and

:args (

{OPEXP

:opno 96

:opfuncid 65

:opresulttype 16

:opretset false

:opcollid 0

:inputcollid 0

:args (

{VAR

:varno 1

:varattno 1

:vartype 23

:vartypmod -1

:varcollid 0

:varnullingrels (b)

:varlevelsup 0

:varreturningtype 0

:varnosyn 1

:varattnosyn 1

:location -1

}

```

    }
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
  }
  :location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXPR
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relic 16561

```

```

:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 118
}
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 130
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 122
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 149
}
{OPEXP
:opno 1676
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 168
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:conststypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 192
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 181
}
)
:location 161
}
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
"l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtkind 0
:relid 16566
:inh true
:relkind r

```

```

:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 19 20)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 270
}
){VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
}
:location 281
}
){OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 308
}
){VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 323
}
}
:location 321
}
)
:location 304
}
)
:mergeActionlist <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 232
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>

```

```

        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location 0
        :stmt_len 0
    }
    :location 208
}
}
:location 145
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
{expr
{VAR
:varno 2
:varattno 1
:varatype 25
:varitypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
{expr
{AGGREF
:aggfnoid 2803
:aggtype 28
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 68
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
}

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTe true
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false

```



```

:inFromCl true
:securityQuals (
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks false
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTY
            :alias
              {ALIAS
                :aliasname c
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname c
                :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
              }
            :rtekind 0
            :relid 16546
            :inh true
            :relkind r
            :rellockmode 1
            :perminfoindex 1
            :tablesample <>
            :lateral false
            :inFromCl true
            :securityQuals <>
          }
        )
      }
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 16546
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 13)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  )
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals
      {BOOLEXP
        :boolop and
        :args (
          {OPEXP
            :opno 96
            :opfuncid 65
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 1
                :location -1
              }
              {VAR
                :varno 1
                :varattno 2
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 1
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 2
                :location -1
              }
            )
          }
        :location -1
      }
      {OPEXP
        :opno 674
        :opfuncid 297
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 6
            :vartype 701
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 6
            :location -1
          }
          {FUNCEXP
            :funcid 316
            :funcresulttype 701
            :funcset false
            :funcvariadic false
            :funcformat 2
            :funccollid 0
            :inputcollid 0
            :args (
              {CONST
                :consttype 23
                :consttypmod -1
                :constcollid 0
                :constlen 4
                :constbyval true
                :constisnull false
                :location -1
              }
            )
          }
        )
      }
    )
  }

```

```

        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
    }
    }
    :location -1
    }
    )
    :location -1
    }
    )
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :resortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
    )
    {RANGETBLENTRY
    :alias <>
    :eref
    {ALIAS
    :aliasname *GROUP*
    :colnames ("o_orderpriority")
    }
    }
    :rtkind 9
    :groupprims (
    {VAR
    :varno 1
    :varattno 6
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location 47
    }
    )
    :lateral false
    :inFromCl false
    :securityQuals <>
    }
    )
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 16561
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 12 13)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    )
    :quals
    {BOOLEXP
    :boolop and
    :args (
    {OPEXP
    :opno 1098
    :opfuncid 1090
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 110
    }
    {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 130
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
    }
    )
    :location 122
    }
    {OPEXP
    :opno 2345

```

```

:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 149
  }
  {OEXPR
    :opno 1076
    :opfuncid 2071
    :opresulttype 1114
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {CONST
        :consttype 1082
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 168
        :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
      }
      {CONST
        :consttype 1186
        :consttypmod 196607
        :constcollid 0
        :constlen 16
        :constbyval false
        :constisnull false
        :location 192
        :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
      }
    )
    :location 181
  }
)
:location 161
}
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:opetName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem
        :colnames ('l_orderkey' 'l_partkey' 'l_suppkey' 'l_linenu
          mber' 'l_quantity' 'l_extendedprice' 'l_discount' 'l_tax'
            'l_returnflag' 'l_linestatus' 'l_shipdate' 'l_commitdate
              'l_receiptdate' 'l_shipinstruct' 'l_shipmode' 'l_commen
                t')
      }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 19 20)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 270
      }
      {VAR
        :varno 1
        :varattno 1
        :vartype 23

```

```

        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 283
    }
}
:location 281
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opreset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 12
        :location 308
    }
    {VAR
        :varno 1
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location 323
    }
}
:location 321
}
}
:location 304
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location 232
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 208
}
}
:location 145
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {VAR
            :varno 2
            :varattno 1
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 1
            :location -1
        }
        :resno 1
        :resname o_orderpriority
        :ressortgroupref 1
        :resorigtbl 16561
        :resorigcol 6
        :resjunk false
    }
}
{TARGETENTRY
:expr
{AGGREF
:aggtfnoid 2803
:aggtype 28
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n

```

```

        :agppresorted false
        :agglevelsup 0
        :aggsplit 0
        :aggn -1
        :aggttransno -1
        :location 68
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
    {SORTGROUPCLAUSE
     :tleSortGroupRef 1
     :eqop 98
     :sortop 664
     :reverse_sort false
     :nulls_first false
     :hashable true
    }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
    {SORTGROUPCLAUSE
     :tleSortGroupRef 1
     :eqop 98
     :sortop 664
     :reverse_sort false
     :nulls_first false
     :hashable true
    }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: ===== FINAL PLAN =====
DETAIL:      {AGG
:plan.disabled_nodes 0
:plan.startup_cost 8883274415.708885
:plan.total_cost 8883274719.763882
:plan.plan_rows 5
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
    {TARGETENTRY
    :expr
    {VAR
     :varno -2
     :varattno 1
     :vartype 25
     :vartypmod -1
     :varcollid 100
     :varnullingrels (b)
     :varlevelsup 0
     :varreturningtype 0
     :varnosyn 0
     :varattnosyn 0
     :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16561
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
     :aggfnoid 2803
     :aggtype 20
     :aggcollid 0
     :inputcollid 0
     :aggttranstype 20
     :aggargtypes <>
     :aggdirectargs <>
     :args <>
     :aggorder <>
     :aggdistinct <>
     :aggfilter <>
     :aggstar true
     :aggvariadic false
     :aggkind n
     :agppresorted false
     :agglevelsup 0
     :aggsplit 0
     :aggn 0
     :aggttransno 0
     :location 68
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    }
)
:plan.qual <>
:plan.lefttree
{SORT
:plan.disabled_nodes 0
:plan.startup_cost 8883274415.708885
:plan.total_cost 8883274517.043884
:plan.plan_rows 40534
:plan.plan_width 9
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
    {TARGETENTRY
    :expr
    {VAR
     :varno -2
     :varattno 1
     :vartype 25
     :vartypmod -1
     :varcollid 100
     :varnullingrels (b)
     :varlevelsup 0
     :varreturningtype 0
     :varnosyn 1
     :varattnosyn 6

```

```

        :location -1
    }
    :resno 1
    :resname <>
    :ressortgrouppref 1
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:plan.qual <>
:plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 212549.3375
:join.plan.total_cost 8883271313.470623
:join.plan.plan_rows 40534
:join.plan.plan_width 9
:join.plan.parallel_aware false
:join.plan.parallel_safe false
:join.plan.async_capable false
:join.plan.plan_node_id 2
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgrouppref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 8883048508
:scan.plan.plan_rows 144199
:scan.plan.plan_width 13
:scan.plan.parallel_aware false
:scan.plan.parallel_safe false
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
:resno 2
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 110
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 130
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 122
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0

```

```

:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramids (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColtype 23
:firstColtypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 5716.52
:per_call_cost 0.0025
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 149
  }
  {CONST
    :consttype 1114
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
  }
)
:location 161
}
}
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 1
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 179731.5
:plan.total_cost 179731.5
:plan.plan_rows 2000307
:plan.plan_width 4
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 4
:plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno -2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 3
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 179731.5
:scan.plan.plan_rows 2000307
:scan.plan.plan_width 4
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 5
:scan.plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 3
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 3
      :varattnosyn 1
      :location 270
    }
    :resno 1
    :resname <>

```

```

:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 1095
:opfuncid 1007
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 308
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 323
}
)
:location 321
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 270
}
)
:skewTable 16561
:skewColumn 1
:skewInherit false
:rows_total 0
}
:join.plan.initPlan <>
:join.plan.extParam (b)
:join.plan.allParam (b)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno -1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 270
}
)
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
)
)
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)

```



```

    }
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 1
:aggssplit 0
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}

-----
QUERY PLAN
-----
GroupAggregate (cost=8883274415.71..8883274719.76 rows=5 width=17) (actual time=2557.906..2559.642 rows=5.00 loops=1)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  Buffers: shared hit=4427 read=129976, temp read=14748 written=14748
  -> Sort (cost=8883274415.71..8883274517.04 rows=40534 width=9) (actual time=2557.454..2558.262 rows=23317.00 loops=1)
    Output: orders.o_orderpriority
    Sort Key: orders.o_orderpriority
    Sort Method: quicksort Memory: 769kB
    Buffers: shared hit=4427 read=129976, temp read=14748 written=14748
    -> Hash Semi Join (cost=212549.34..8883271313.47 rows=40534 width=9) (actual time=1740.926..2550.157 rows=23317.00 loops=1)
      Output: orders.o_orderpriority
      Hash Cond: (orders.o_orderkey = lineitem.l_orderkey)
      Buffers: shared hit=4424 read=129976, temp read=14748 written=14748
      -> Seq Scan on public.orders (cost=0.00..8883048508.00 rows=144199 width=13) (actual time=41.682..403.097 rows=25383.00 loops=1)
        Output: orders.o_orderpriority, orders.o_orderkey
        Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).col1)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))
      ne))
        Rows Removed by Filter: 1474617
        Buffers: shared hit=4048 read=25632
        SubPlan 2
        -> Seq Scan on public.customer c (cost=0.00..5547.00 rows=67808 width=4) (actual time=0.013..28.347 rows=67989.00 loops=1)
          Output: c.c_custkey
          Filter: (c.c_acctbal > '5000'::double precision)
          Rows Removed by Filter: 82011
          Buffers: shared hit=3672
        -> Hash (cost=179731.50..179731.50 rows=2000307 width=4) (actual time=1694.607..1694.608 rows=3793296.00 loops=1)
          Output: lineitem.l_orderkey
          Buckets: 262144 (originally 262144) Batches: 32 (originally 16) Memory Usage: 6223kB
          Buffers: shared hit=376 read=104344, temp written=10750
          -> Seq Scan on public.lineitem (cost=0.00..179731.50 rows=2000307 width=4) (actual time=0.615..1133.957 rows=3793296.00 loops=1)
            Output: lineitem.l_orderkey
            Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)
            Rows Removed by Filter: 2207919
            Buffers: shared hit=376 read=104344

Planning:
  Buffers: shared hit=219
Planning Time: 4.771 ms
Execution Time: 2561.848 ms
(37 rows)

tpch=>

```